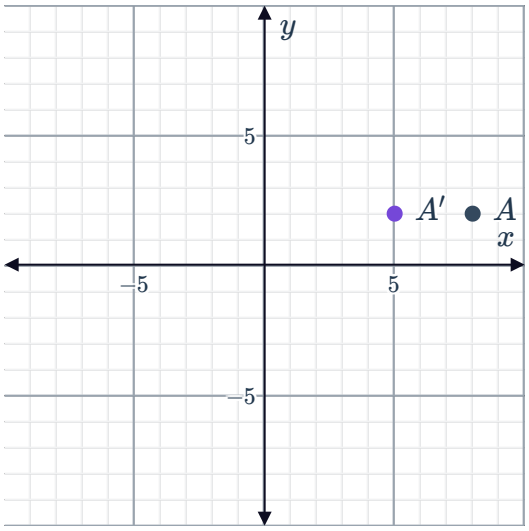
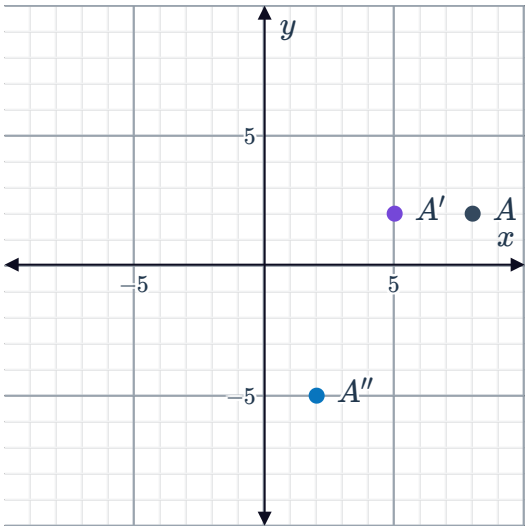
Compositions of transformations

1

a

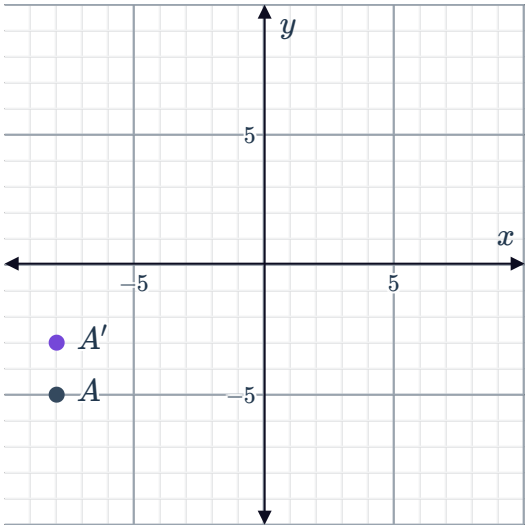


b

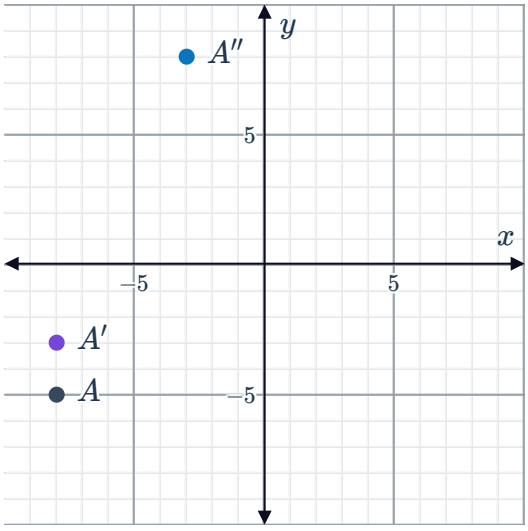


2

a



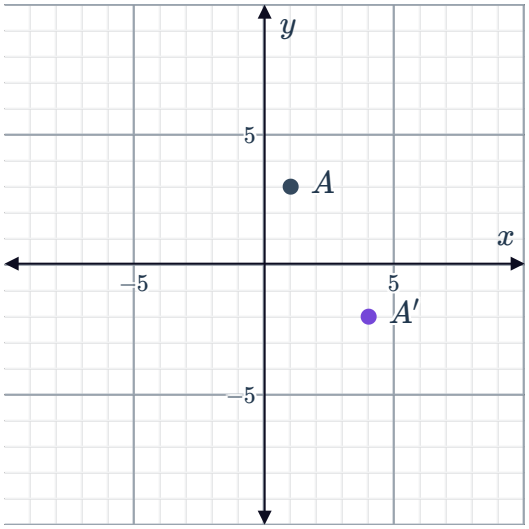
b



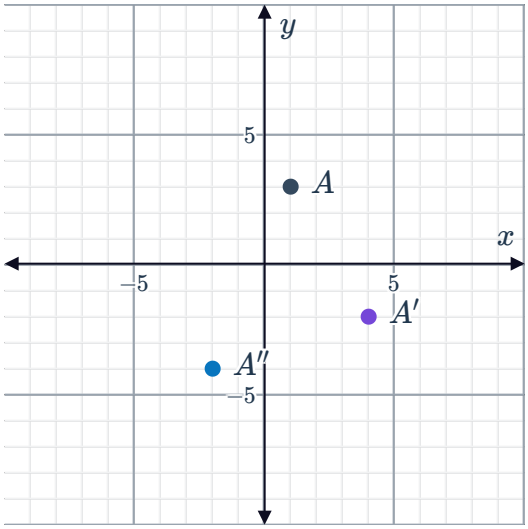
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3

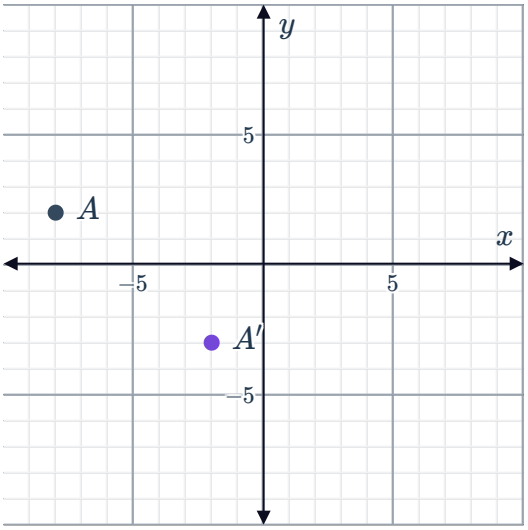
a



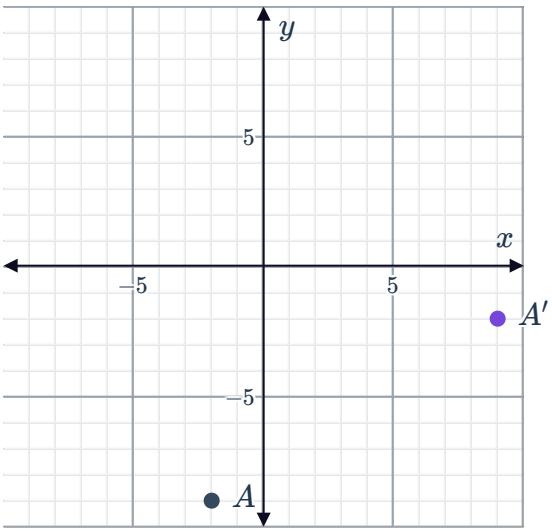
b



4

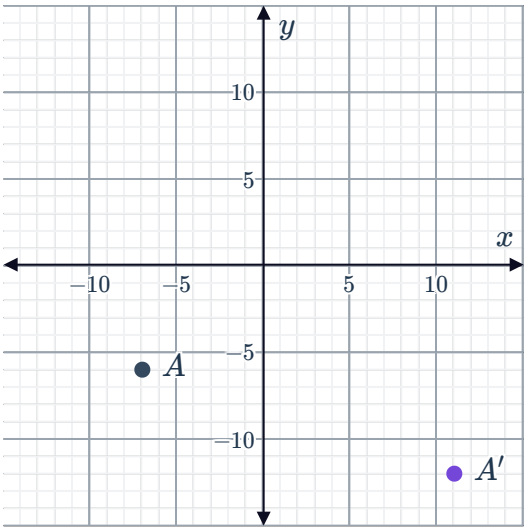


5



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6



7

a $B\,(-4,-4)$

b $y = 0$

8

a $(-a,-b)$

b True

9

a $B\,(7,9)$

b 16 units to the left

10

a $(-3,9)$

b $(-4,9)$

c $(3,-4)$

d $(-10,3)$

11

a $A'\,(-6,-1)$

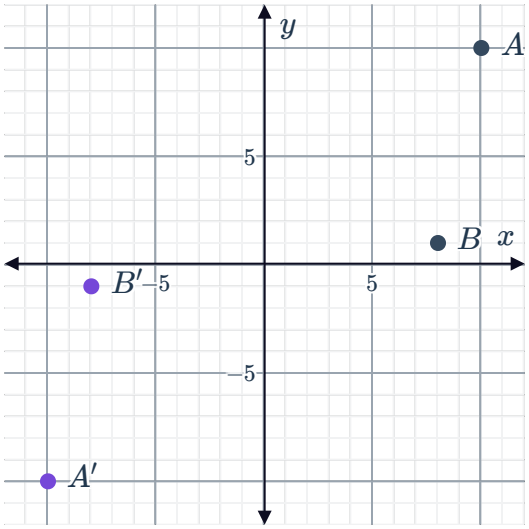
b

i Yes **ii** No **iii** No **iv** No

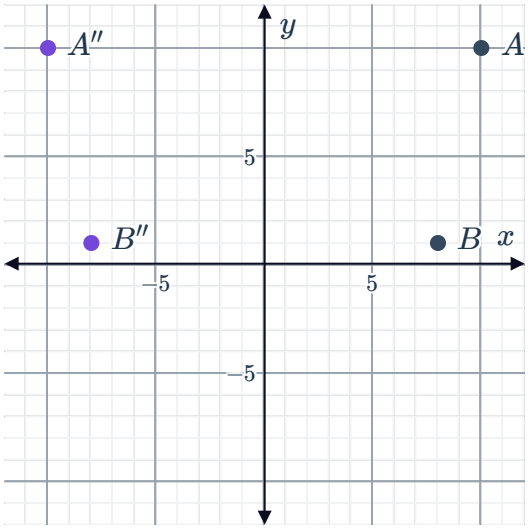
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12

a



b

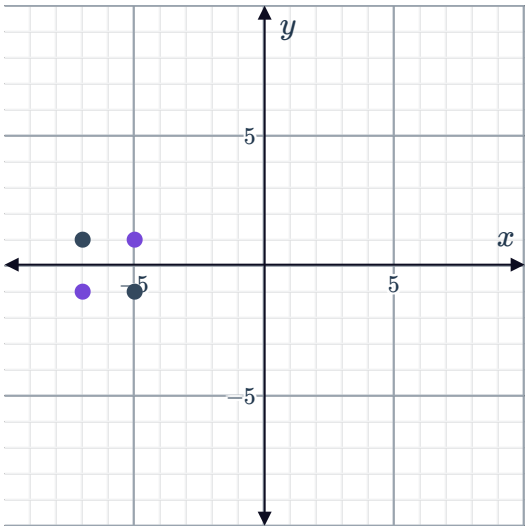


c

i No **ii** Yes **iii** No **iv** No

13

a



b

i No **ii** No **iii** No **iv** Yes

14

a $A'(-10, 8)$
$B'(-7, 7)$

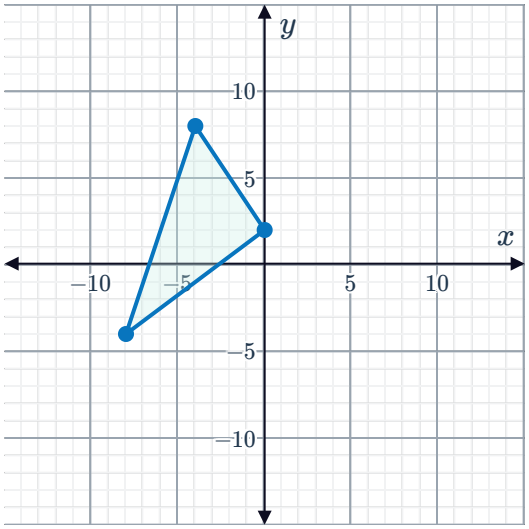
b

i No **ii** Yes **iii** No **iv** No

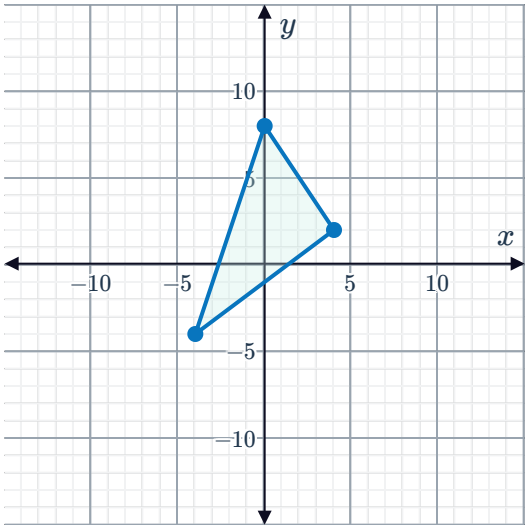
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15

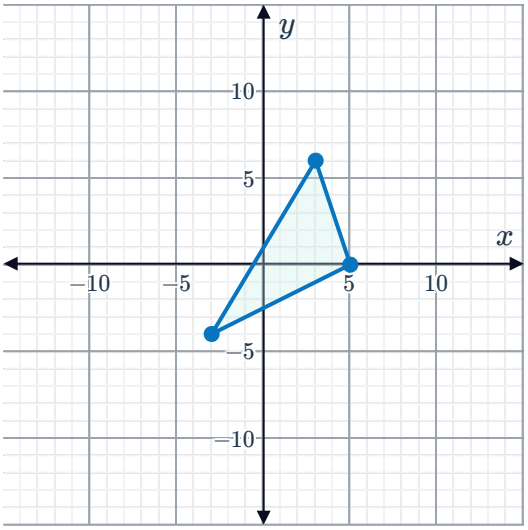
a



b

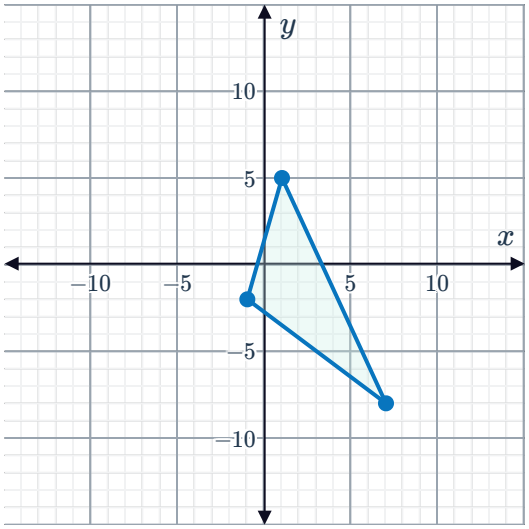


16

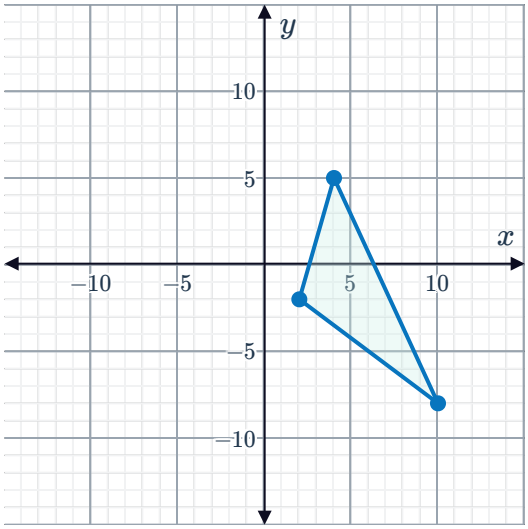


17

a

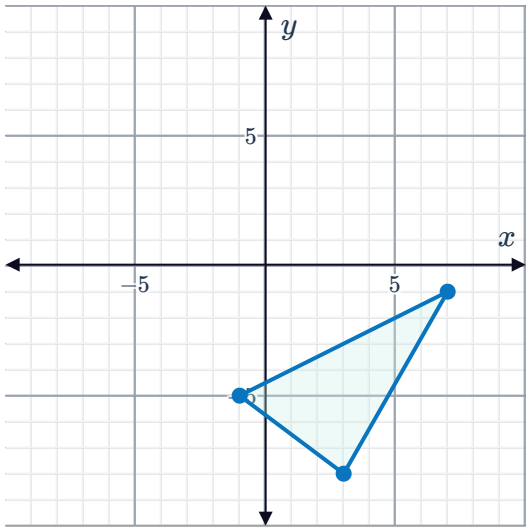


b



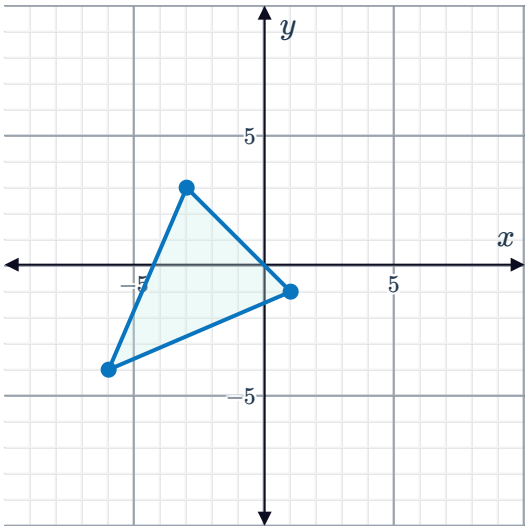
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18

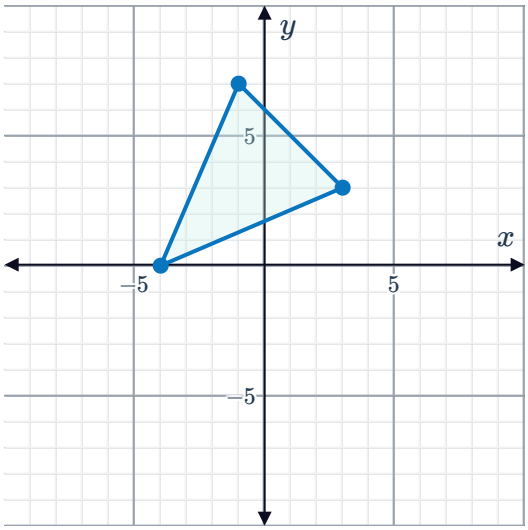


19

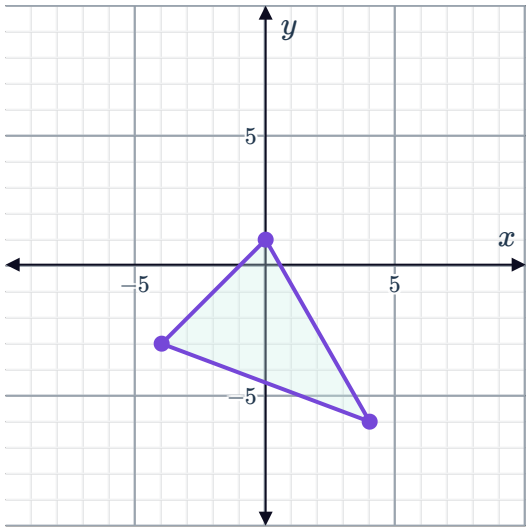
a



b



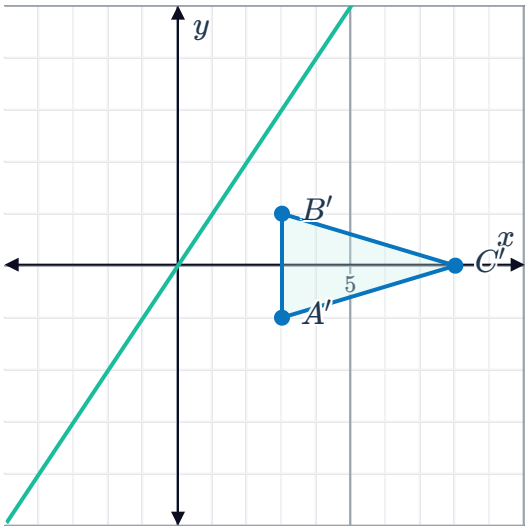
20



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21

a



b Possible answers:
A 90° rotation counterclockwise about the origin.
A 270° rotation clockwise about the origin.

22 $A'(-3,-5)$
$B'(-1,8)$
$C'(14,10)$

23 $A'(-10,-7)$
$B'(5,2)$
$C'(5,-1)$

24 $A'(1,-9)$
$B'(1,3)$
$C'(13,2)$

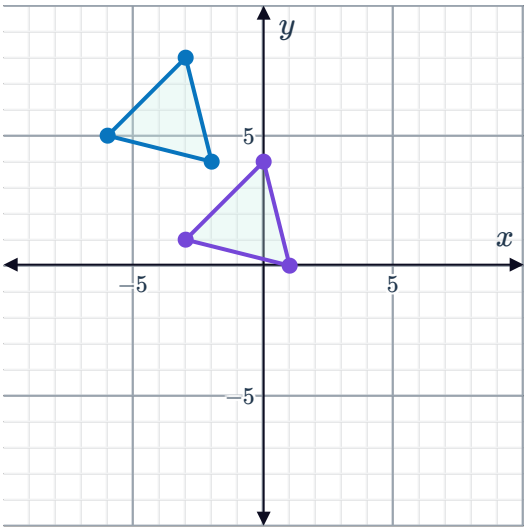
25 $A'(-8,-1)$
$B'(-5,9)$
$C'(-9,12)$

26 $A'(4,6)$
$B'(-1,5)$
$C'(13,-12)$

27 $A'(8,-2)$
$B'(-1,9)$
$C'(-6,6)$

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28



29

a No **b** No **c** No **d** Yes

30

a No **b** No **c** No **d** Yes

31

a No **b** No **c** Yes **d** No

32

a

$A(-2,2)$	$\rightarrow$	$A'(3,-2)$
$B(4,-3)$	$\rightarrow$	$B'(9,-7)$
$C(-4,-3)$	$\rightarrow$	$C'(1,-7)$

b

$A'(3,-2)$	$\rightarrow$	$A''(2,3)$
$B'(9,-7)$	$\rightarrow$	$B''(7,9)$
$C'(1,-7)$	$\rightarrow$	$C''(7,1)$

33

a

$A(-7,-2)$	$\rightarrow$	$A'(-2,7)$
$B(-3,-2)$	$\rightarrow$	$B'(-2,3)$
$C(-3,-8)$	$\rightarrow$	$C'(-8,3)$
$D(-7,-8)$	$\rightarrow$	$D'(-8,7)$

b

$A'(-2,7)$	$\rightarrow$	$A''(2,7)$
$B'(-2,3)$	$\rightarrow$	$B''(2,3)$
$C'(-8,3)$	$\rightarrow$	$C''(8,3)$
$D'(-8,7)$	$\rightarrow$	$D''(8,7)$

34

a No **b** Yes **c** No **d** No

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35
a No **b** No **c** Yes **d** No

36
a No **b** No **c** No **d** Yes

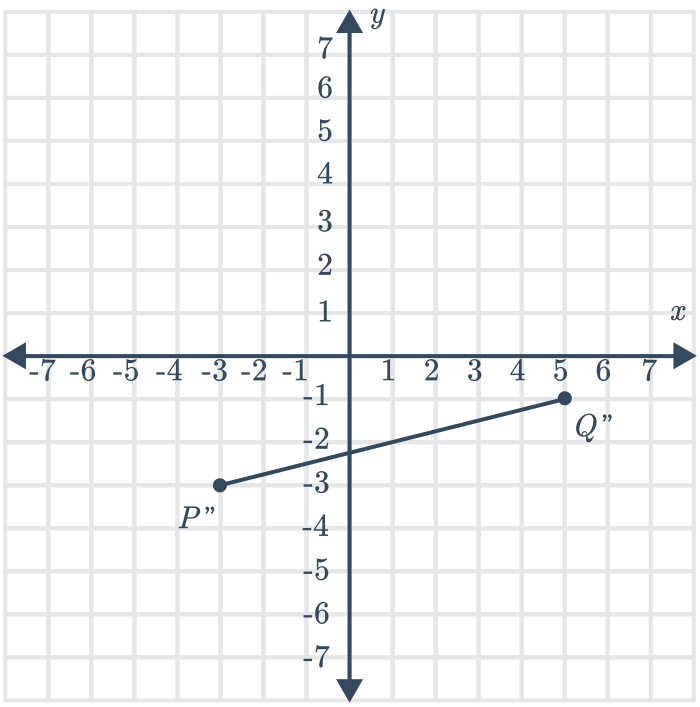
37
a No **b** No **c** No **d** Yes

38
a Yes **b** No **c** No **d** No

39 $(18, -18), (18, 0), (0, 0), (0, -18)$

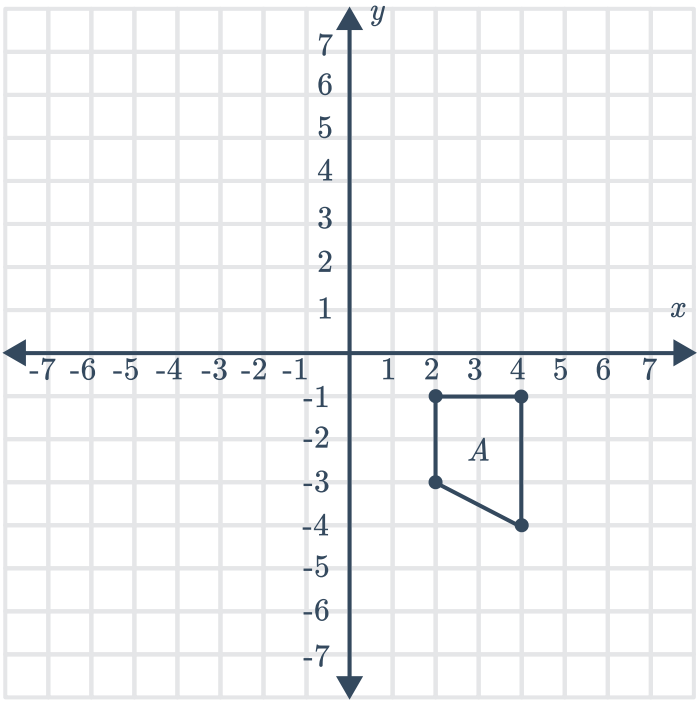
Additional questions

40



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41

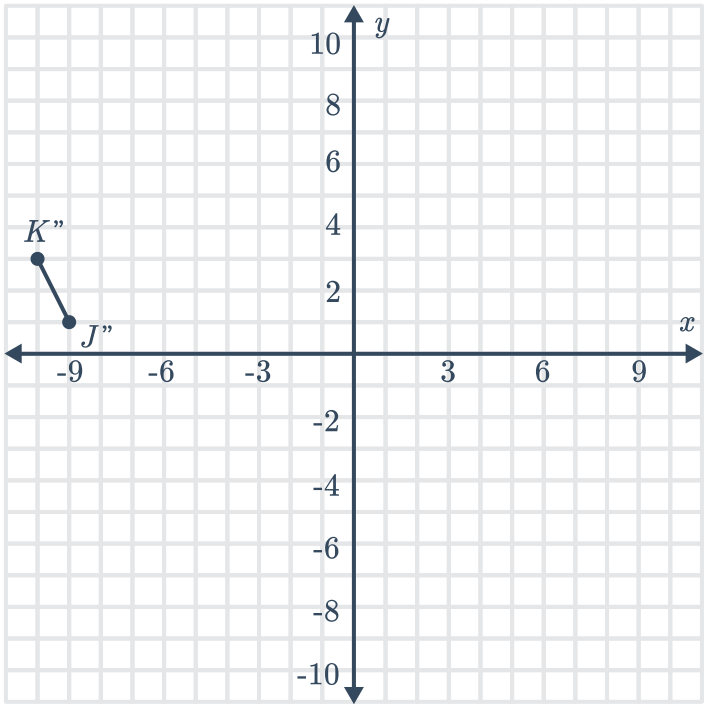


42 Reflect $\triangle ABC$ across the x-axis, then translate $\triangle A'B'C'$ by 2 units up and 6 units to the right.

43 Reflect $\triangle ABC$ across the x-axis, then rotate $\triangle A'B'C'$ by 90° counterclockwise about the origin.

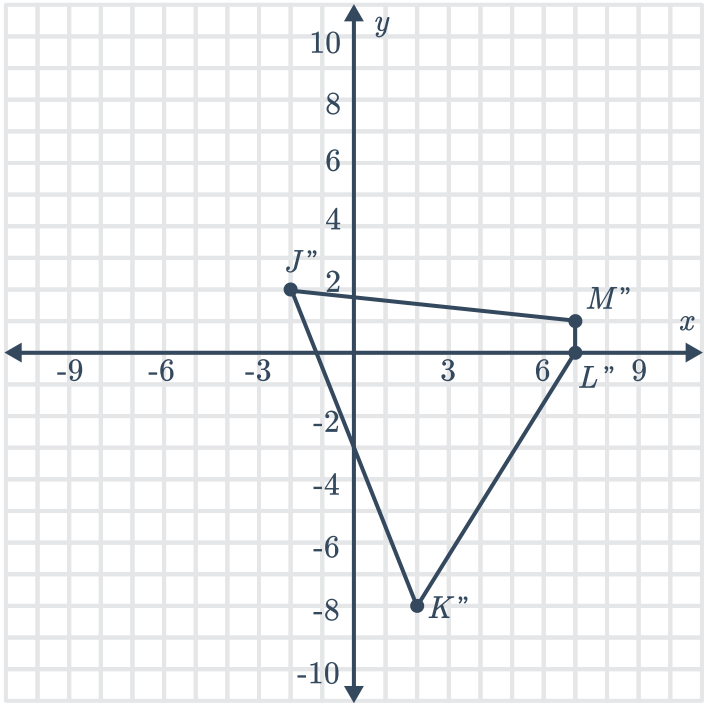
44 Translate 1 unit down then reflect $\triangle ABC$ across the line $y = x$.

45



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46



47

a Rotate section 1 180° clockwise or counterclockwise about the center of the flag.

b Reflect section 1 over the horizontal midline, then the vertical midline (in either order).

48 Answers may vary. Starting with the top leftmost parallelogram in the pattern, perform a series of translations down the fabric until you have six parallelograms. Then, reflect the set of six parallelograms across the line of symmetry in the pattern.

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